

BAULKHAM HILLS HIGH SCHOOL

YEAR 12

HALF YEARLY EXAMINATION

2007

MATHEMATICS EXTENSION 1

GENERAL INSTRUCTIONS:

- Attempt ALL questions.
- Start each of the 7 questions on a new page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet.
- Marks indicated for each question are only a guide and could change.

2

QUESTION 1

Marks

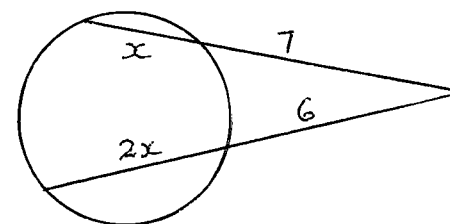
- (a) Solve $2\log x = \log 6x$ 2
- (b) (i) Differentiate $\log_e (\cos x)$ 1
- (ii) Hence evaluate $\int_0^{\frac{\pi}{3}} \tan x \, dx$ 3
- (c) Given the points A(-1,4) and B(2,-3), find the coordinates of the point P(x,y) which divides the interval AB externally in the ratio 2 : 3. 3
- (d) Solve $\frac{x}{x+2} \geq 4$. 3

QUESTION 2

- (a) Solve for $0 \leq \theta \leq 360^\circ$:
- (i) $\sin 2\theta - \cos \theta = 0$. 3
- (ii) $\sin \theta + \sqrt{3} \cos \theta = 1$. 3
- (b) (i) Show that the derivative of $y = \sec x$ is $\tan x \sec x$. 3
- (ii) Hence find the equation of the tangent to the curve $y = \sec x$ at $\frac{\pi}{6}$. 3

QUESTION 3

- (a) Find x in the following:

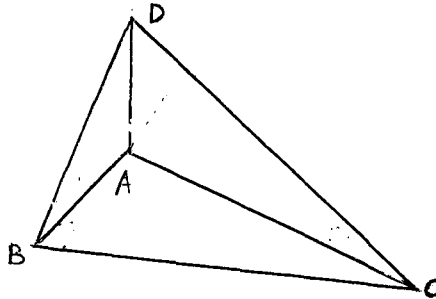


- (b) Find the volume of the solid generated when the area between the curve $y = \sin x$ the x axis and the lines $x = 0$ and $x = \frac{\pi}{4}$ is rotated about the x axis. 3
- (c) The polynomial $P(x) = 2x^3 + ax + b$ has a root at $x = 1$ and when divided by $x + 2$ the remainder is 4. Find the values of a and b. 3

QUESTION 3 (Continued)

Marks

- (d) From a point B on a wharf it is noted that the cruise liner Queen Mary II bears due north and its flagpole at D has an angle of elevation of 17° . From another point C 280 metres due east of B the angle of elevation to the flagpole is 12° .



- (i) Show that the height (h) of the flagpole above the wharf is given by:

$$h = 280 (\cot^2 12^\circ - \cot^2 17^\circ)^{-1/2}$$

- (ii) Find h the height of the flagpole above the wharf.

3

1

QUESTION 4

- (a) Find $\int \sin x \cos^3 x \, dx$.
- (b) If $\log_a 3 = x$ and $\log_a 4 = y$, find $\log_a 6$ in terms of x and y .
- (c) Prove by mathematical induction that $7^n - 3^n$ is divisible by 4 for all positive integers n where $n \geq 1$.
- (d) Use Newton's method once to find a better approximation for the root of $f(x) = \ln x - x^3 + 2$. Let $x = 2$ be the first estimate for the root.
- (e) Find the exact value of $\cos 75^\circ$.

2

2

3

3

2

QUESTION 5

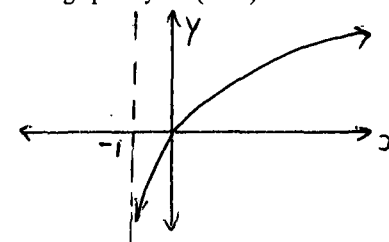
Marks

- (a) TP is a tangent at P.

Prove TRBP is a cyclic quadrilateral.

3

- (b) Below is the graph of $y = \ln(x + 1)$.



- (i) Find the gradient of the tangent to the curve $y = \ln(x + 1)$ at the origin.
- (ii) Hence find the acute angle between $y = \ln(x + 1)$ and the line $y = 4x$ at $x = 0$.
- (iii) For what range of values for m (where $m > 0$) will $mx - \ln(x + 1) = 0$ have 2 solutions.

1

2

1

- (c) When the temperature T of a certain body is 65°C , it is cooling at a rate of 1°C per minute. Assuming Newton's law of cooling:

$$\frac{dT}{dt} = -k(T - S),$$

where T is the temperature of the body at time t minutes and S is that of the surrounding medium and k is a constant.

- (i) Verify that $T = S + Ae^{-kt}$ is a solution of the differential equation above.
- (ii) Show that $k = 0.02$ given that $S = 15^\circ\text{C}$.
- (iii) Find the temperature of the body after 20 minutes.
- (iv) How much more time must elapse before the temperature of the body reaches 35°C .

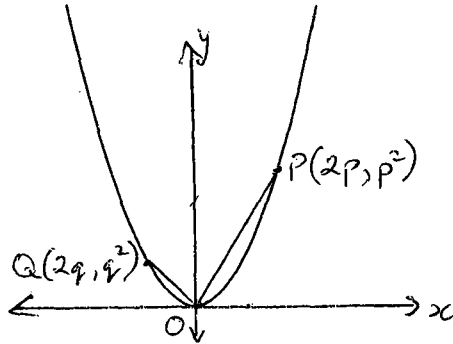
1

1

1

2

QUESTION 6



Marks

- (a) (i) What is the cartesian equation represented by the parametric equations:
 $x = 2t$ $y = t^2$ 1
- (ii) For 2 points $P(2p, p^2)$ and $Q(2q, q^2)$ on the parabola the chord joining them subtends an angle of 90° at the origin (i.e. $\angle QOP = 90^\circ$). By finding the gradients of OP and OQ show that $pq = -4$. 2
- (iii) Show clearly that the coordinates of R such that OPRQ is a rectangle is given by $(2p + 2q, p^2 + q^2)$. 2
- (iv) Find the locus of R as P and Q move on the parabola such that OPRQ is always a rectangle. 2
- (b) (i) Sketch $y = \frac{1}{|x - 2|}$ without using calculus. 2
- (ii) Hence or otherwise solve $\frac{1}{|x - 2|} < 1$. 2

QUESTION 7

- (a) If $f(x) = -x^2 e^{-x}$
- (i) Find the stationary points on the curve and determine their nature. 4
- (ii) Sketch the curve. 2
- (iii) If $f''(x) = -e^{-x}(x^2 - 4x + 2)$ for what value(s) of x will $f'(x) > 0$ and $f''(x) < 0$. 3
- (b) (i) Factorise $a^3 - b^3$. 1
- (ii) Hence show $\lim_{x \rightarrow 0} \frac{(2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}}}{x} = \frac{\sqrt[3]{2}}{6}$ 3

Question 1 - 12 marks

1a) $2 \log x = \log 6x$
 $\therefore \log x^2 = \log 6x$
 $x^2 = 6x \dots \dots \textcircled{1}$
 $x(x-6) = 0$
 $\therefore x = 0, 6$ but $x > 0$
 $\therefore x = 6 \dots \dots \textcircled{1}$

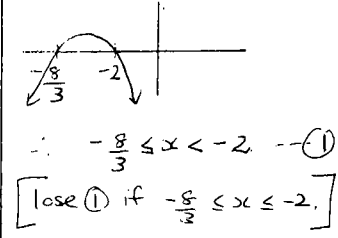
b) (i) $y = \log_e(\cos x)$
 $y' = \frac{-\sin x}{\cos x}$
 $= -\tan x \dots \dots \textcircled{1}$

(ii) $\int_0^{\frac{\pi}{3}} \tan x = [\log_e(\cos x)]_0^{\frac{\pi}{3}}$
 $= -\log_e(\cos \frac{\pi}{3}) + \log_e(\cos 0)$
 $= -\log_e(\frac{1}{2}) + \log_e 1$
 $= -\log(\frac{1}{2})$
 $= \log 2$
 $= 0.693 \dots \dots \textcircled{1}$

c) A(-1,4) B(2,-3)
 $\textcircled{3} \textcircled{1} \leftarrow -2:3$
 $\left(\frac{-2(2)+3(-1)}{-2+3}, \frac{-2(-3)+3(4)}{-2+3} \right)$
 $= (-7, 18) \dots \dots \textcircled{1}$

d) $\frac{x}{x+2} \geq 4 \quad x \neq -2$
 $\frac{x}{x+2} - 4 \geq 0$
 $\frac{x - 4(x+2)}{x+2} \geq 0$
 $\frac{x - 4x - 8}{x+2} \geq 0$
 $\frac{-3x - 8}{x+2} \geq 0$
 $(x+2)(-3x-8) \geq 0 \dots \dots \textcircled{1}$

$\frac{\pi}{3} (x+2)(-3x-8) \geq 0$
 $(x+2)(-3x-8) \geq 0 \dots \dots \textcircled{1}$



[lose 1 if $-\frac{8}{3} \leq x \leq -2$]

Question 2 - 12 marks

a) (i) $\sin 2\theta - \cos \theta = 0$
 $2 \sin \theta \cos \theta - \cos \theta = 0$
 $\cos \theta (2 \sin \theta - 1) = 0$
 $\cos \theta = 0 \quad \sin \theta = \frac{1}{2}$
 $\theta = 90^\circ, 270^\circ; 30^\circ, 150^\circ$

(ii) $\textcircled{1} \quad \textcircled{1}$
 $\sin \theta + \sqrt{3} \cos \theta = 1$
Let $A \sin(\theta + \alpha) = \sin \theta + \sqrt{3} \cos \theta$
 $A \sin \alpha \cos \theta + A \cos \alpha \sin \theta = \sin \theta + \sqrt{3} \cos \theta$
 $A \cos \alpha = 1 \quad A \sin \alpha = \sqrt{3}$
 $\tan \alpha = \sqrt{3} \quad A = \sqrt{1+3}$
 $\alpha = 60^\circ \quad A = 2 \dots \dots \textcircled{1}$

$\therefore 2 \sin(\theta + 60^\circ) = 1$
 $\sin(\theta + 60^\circ) = \frac{1}{2}$
 $\theta + 60^\circ = 30^\circ, 150^\circ$
 $\theta = -30^\circ, 90^\circ$
 $\theta = 330^\circ, 90^\circ \dots \dots \textcircled{1}$

$\therefore k = 280$
 $\sqrt{(\cot^2 12^\circ - \cot^2 17^\circ)}$

i) $k = 280 (\cot^2 12^\circ - \cot^2 17^\circ)^{-\frac{1}{2}}$
(ii) $82.8 \text{ m} \dots \dots \textcircled{1}$

Question 4 - 12 marks

a) $\int \sin x \cos^3 x \, dx$
 $\frac{d}{dx} (\cos x)^4 = -4 \sin x (\cos x)^3$
 $\therefore \int \sin x \cos^3 x \, dx = -\frac{1}{4} \cos^4 x + C \dots \dots \textcircled{1}$

b) $\log_a 3 = x \quad \log_a 4 = y$
 $\log_a 6 = \log_a 3 \times \sqrt{4} \dots \dots \textcircled{1}$
 $= \log_a 3 + \log_a 2$
 $= \log_a 3 + \frac{1}{2} \log_a 4$
 $= x + \frac{y}{2} \dots \dots \textcircled{1}$

c) $1 - 2 \dots \dots \textcircled{1}$

Step 1: prove true for $n=1$
 $7^1 - 3^1 = 4 \checkmark \dots \dots \textcircled{1}$

Step 2: Assume true for $n=k$ ie $7^k - 3^k = M$
ie $7^k = 4M + 3^k$ (integer)
 $\therefore 7^{k+1} = 7(4M + 3^k) = 28M + 7(3^k)$
 $7^{k+1} - 3^{k+1} = 28M + 7(3^k) - 3(3^k)$
 $= 28M + 4(3^k)$
 $= 4(7M + 3^k)$
which is $\div$ by 4.

Step 3: Prove true for $n=k+1$
ie $7^{k+1} - 3^{k+1} = 4(7M + 3^k)$
Step 4: Proved true for $n=1$
assume true for $n=k$
proven true for $n=k+1$
 $\therefore$ If true for $n=1$ true for $n=2, n=3$ & for all n by mathematical ind.

a) $a_1 = 2 - \frac{f(2)}{f'(2)}$

$f(x) = \frac{1}{x} - 3x^2$
 $f(2) = \ln 2 - 8 + 2 = \ln 2 - 6$
 $f'(x) = -\frac{1}{x^2} - 6x$
 $f'(2) = -\frac{1}{4} - 12 = -11.5$
 $a_1 = \frac{2 - \ln 2}{-11.5} = 1.54 \dots \dots \textcircled{1}$

e) $\cos(75^\circ) = \cos(45^\circ + 30^\circ)$
 $= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$
 $= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$
 $= \frac{\sqrt{3}-1}{2\sqrt{2}}$
or $\frac{\sqrt{6}-\sqrt{2}}{4} \dots \dots \textcircled{1}$

2 b) (i) $y = \sec x$

$y = (\cos x)^{-1} \dots \dots \textcircled{1}$
 $y' = -1(\cos x)^{-2} \cdot -\sin x$
 $= \frac{\sin x}{\cos^2 x}$
 $= \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x}$
 $= \tan x \cdot \sec x$

(ii) $y = \sec x$
at $x = \frac{\pi}{6} \quad \sec \frac{\pi}{6} = \frac{2}{\sqrt{3}} \dots \dots \textcircled{1}$
at $x = \frac{\pi}{6} \quad y' = \tan \frac{\pi}{6} \sec \frac{\pi}{6}$
 $= \frac{1}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}}$
 $= \frac{2}{3} \dots \dots \textcircled{1}$

Equation of tangent
 $y - \frac{2}{\sqrt{3}} = \frac{2}{3} (x - \frac{\pi}{6})$
 $y = 2x - \frac{\pi}{3} + \frac{2}{3} \dots \dots \textcircled{1}$

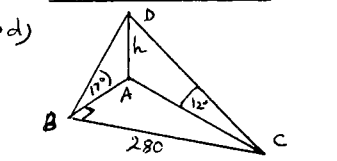
Question 3 - 12 marks

a) $7(7+x) = 6(2x+6) \dots \dots \textcircled{1}$
 $49 + 7x = 12x + 36$
 $13 = 5x$
 $x = \frac{13}{5} \dots \dots \textcircled{1}$

b) $V = \pi \int_0^{\frac{\pi}{4}} \sin^2 x \, dx$
 $V = \pi \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{2} \, dx$
 $= \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}}$
 $= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{\sin \frac{\pi}{2}}{2} - 0 \right]$
 $= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right]$
 $= \frac{\pi^2}{8} - \frac{\pi}{4} \dots \dots \textcircled{1}$
(or $0.448 \dots$)

c) $P(x) = 2x^3 + ax + b$

$\textcircled{3} P(1) = 0$
 $\therefore 2 + a + b = 0 \dots \dots \textcircled{A}$
 $P(-2) = 4$
 $\therefore -16 - 2a + b = 4$
 $-2a + b = 20 \dots \dots \textcircled{B}$
Solving $\textcircled{A} + \textcircled{B}$ simult.
 $a = -\frac{22}{3}, b = \frac{16}{3} \dots \dots \textcircled{1}$



3d) $\cot 17^\circ = \frac{AB}{h} \rightarrow h \cot 17^\circ = AB$
 $\cot 12^\circ = \frac{AC}{h} \rightarrow AC = h \cot 12^\circ$
 $h^2 \cot^2 12^\circ = h^2 \cot^2 17^\circ + 280^2$
 $h^2 (\cot^2 12^\circ - \cot^2 17^\circ) = 280^2$

Question 5 - 12 marks

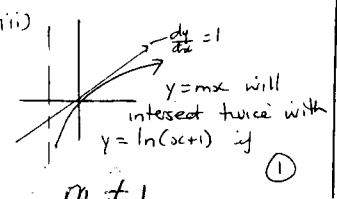
a) Let $\angle AQT = x^\circ$
 $\therefore \angle ABR = x^\circ$ (Exterior)
 $\angle$ of a cyclic quad = interior opposite $\angle$
Let $\angle PAQ = y^\circ$
 $\therefore \angle PBA = y^\circ$ (Angle between tangent & chord = angle in alternate segment)
 $\therefore \angle PBR = x^\circ + y^\circ$
 $\angle PTQ = 180 - (x+y)$
Angle sum of Δ $\dots \dots \textcircled{1}$

$\therefore \angle PBR + \angle PTQ = x+y + 180 - (x+y) = 180^\circ$
 $\therefore$ TPBR is a cyclic

quad as opposite $\angle$'s are supplementary (lose 1 if no conclusion)

b) i) $y = \ln(x+1)$
 $y' = \frac{1}{x+1}$
at $x=0 \quad y' = \frac{1}{0+1}$
 $y' = 1 \dots \dots \textcircled{1}$

(ii) $m_1 = 1 \quad m_2 = 4$
 $\tan \alpha = \frac{|m_1 - m_2|}{1 + m_1 m_2}$
 $= \frac{|1 - 4|}{1 + 4 \times 1}$
 $= \frac{3}{5}$
 $\alpha = 30^\circ 58' \dots \dots \textcircled{1}$



c) $\frac{dT}{dt} = -k(T-S) \dots \dots \textcircled{A}$
 $T = S + Ae^{-kt} \dots \dots \textcircled{B}$

show $\textcircled{B}$ is a solution to $\textcircled{A}$
 $\frac{dT}{dt} = -kAe^{-kt}$
from $\textcircled{B} \quad T-S = Ae^{-kt}$
 $\therefore \frac{dT}{dt} = -k(T-S)$
 $\therefore$ it is a solution

(ii) when $T = 65 \quad \frac{dT}{dt} = -1$
 $-1 = -k(65-15)$
 $k = \frac{1}{50}$
 $k = 0.02$

(iii) when $T = 20$
 $T = 15 + Ae^{-0.02t}$
 $65 = 15 + Ae^0$
 $A = 50$
 $T = 15 + 50e^{-0.02t}$
 $= 48.5^\circ \text{C}$

(1r) find t when $l = 35$

$$35 = 15 + 50e^{-0.02t}$$

$$\frac{20}{50} = e^{-0.02t}$$

$$\therefore \ln(0.4) = -0.02t$$

$$t = \frac{\ln(0.4)}{-0.02} \quad \text{①}$$

$$\therefore A \text{ further } 25.8 \text{ mins.} \quad \text{①}$$

Question 6 - 11 marks

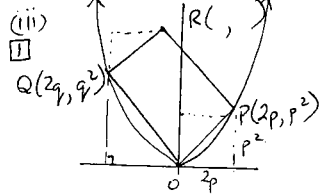
a) (i) $x = 2t$ $y = t^2$

① $t = \frac{x}{2} \therefore y = \left(\frac{x}{2}\right)^2$ ①
 $y = \frac{x^2}{4}$
 $x^2 = 4y$

(ii) $m_{op} = \frac{p^2 - 0}{2p - 0} = \frac{p}{2}$ ①
 similarly $m_{oa} = \frac{q}{2}$

$m_{op} \times m_{oa} = -1 \therefore \frac{p}{2} \times \frac{q}{2} = -1$ ①

$$\therefore pq = -4$$



From a $\uparrow p^2$ then $\rightarrow 2p$ (could use midpoint)
 ie $R(2q+2p, q^2+p^2)$ ①

(iv) $x = 2p + 2q$ $y = p^2 + q^2$
 $\therefore p + q = \frac{x}{2}$ ① $y = (p+q)^2 - 2pq$

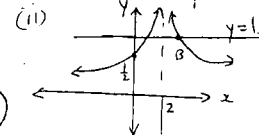
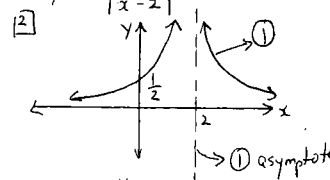
$$\therefore y = \left(\frac{x}{2}\right)^2 - 2(-4) \quad \text{①}$$

$$y = \frac{x^2}{4} + 8$$

$$4y = x^2 + 32$$

$$x^2 - 4y + 32 = 0$$

b) $y = \frac{1}{|x-2|}$



need A & B
 at A $\frac{1}{x-2} = 1$
 $-1 = x-2$
 $x = 1$

at B $\frac{1}{x-2} = 1$
 $x-2 = 1$
 $x = 3$

$\therefore \frac{1}{|x-2|} < 1$

$\therefore x < 1 \quad x > 3$
 ① ①

7a) Question 7 - 13 marks

$$f(x) = -x^2 e^{-x}$$

(i) $f'(x) = -2x e^{-x} + -e^{-x} \cdot (-x^2)$
 $= -2x e^{-x} + x^2 e^{-x}$
 $= -x e^{-x} (2-x)$ ①

$x = 0$ $x = 2$
 $y = 0$ $y = -\frac{4}{e^2}$ ①

test

$x = 0$

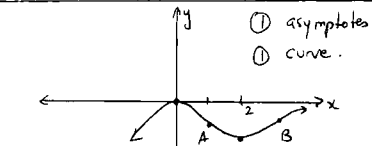
x	-1	0	1
f'(x)	e^{-1}	0	$-1e^{-1}$
	↗	↔	↘

$\therefore$ Max.

test $x = 2$

x	1	2	3
f'(x)	e^{-1}	0	$-2e^{-1}$
	↗	↔	↘

$\therefore$ Min.



(ii) $f''(x) = -e^{-x} (x^2 - 4x + 2)$
 $f''(x) = 0$ when $x^2 - 4x + 2 = 0$
 $\therefore x = \frac{4 \pm \sqrt{16 - 4 \times 1 \times 2}}{2 \times 1}$
 $= \frac{4 \pm 2\sqrt{2}}{2}$
 $= 2 \pm \sqrt{2}$ ①

at A $x = 2 - \sqrt{2}$ at B $x = 2 + \sqrt{2}$

$f'(x) > 0$ when $f(x)$ is increasing
 $f''(x) < 0$ when $f(x)$ is concave down

This occurs when

$x < 0$ and $x > 2 + \sqrt{2}$

① ①
 b) $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$ ①

(i) $a - b = \frac{a^3 - b^3}{a^2 + ab + b^2}$

let $a = (2+x)^{\frac{1}{3}}$ and $b = 2^{\frac{1}{3}}$

$\therefore (2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}} = \frac{(2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}}}{(2+x)^{\frac{2}{3}} + (2+x)^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} + 2^{\frac{2}{3}}}$ ①

$(2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}} = \frac{x}{(2+x)^{\frac{2}{3}} + (2+x)^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} + 2^{\frac{2}{3}}}$

$\therefore (2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}} = \frac{1}{(2+x)^{\frac{2}{3}} + (2+x)^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} + 2^{\frac{2}{3}}}$

$\therefore \lim_{x \rightarrow 0} \frac{(2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}}}{x} = \frac{1}{(2+0)^{\frac{2}{3}} + (2+0)^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} + 2^{\frac{2}{3}}}$

$= \lim_{x \rightarrow 0} \frac{1}{(2+x)^{\frac{2}{3}} + (2+x)^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} + 2^{\frac{2}{3}}}$

$= \frac{1}{(2+0)^{\frac{2}{3}} + (2+0)^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} + 2^{\frac{2}{3}}}$

$= \frac{1}{2^{\frac{2}{3}} + 2^{\frac{1}{3}} + 2^{\frac{2}{3}}}$

$= \frac{1}{3 \times 2^{\frac{2}{3}}}$

$= \frac{1}{3} \times 2^{-\frac{2}{3}} \times 2 = \frac{1}{3} \times 2^{\frac{1}{3}} \times 2 = \frac{1}{3} \times 2^{\frac{4}{3}}$ ①

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$